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NETWORK OBSERVABILITY POWERED BY KENTIK

Kentik Deployment Guide

Prepared for Northwind Health

Data center and AWS visibility

Customer	Northwind Health
Project	Data center and AWS visibility
Prepared by	Infoblox Solutions Engineering
Date	2026-10-29
Kentik region	United States (Ashburn, VA)
Scope	1 data center(s), AWS, Synthetic monitoring

This guide describes the Kentik deployment designed for this environment: what gets installed, which permissions and firewall rules are needed, and the steps to go live. Facts are taken from the Kentik Knowledge Base (kb.kentik.com) and each section names its sources.

Infoblox Confidential. Prepared for the named customer only.

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1. Deployment at a glance

Kentik is SaaS. The customer runs a small footprint where the network lives, and everything lands in the customer's Kentik account in the United States (Ashburn, VA) region over encrypted, outbound connections.

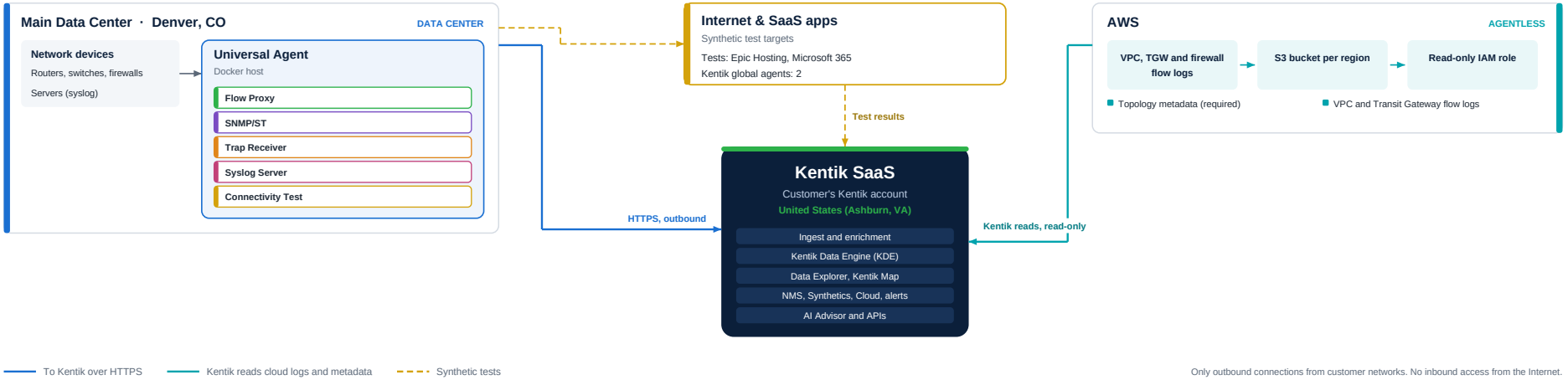
Where	Deployment	What Kentik collects	Footprint
Main Data Center (Denver, CO)	Universal Agent on Docker	Flow Proxy: Forwarding, SNMP/ST, SNMP Trap Receiver, Syslog Server, Connectivity Test	17 vCPU / 56 GB
Amazon Web Services	Agentless (read-only access)	Topology metadata (required), VPC and Transit Gateway flow logs	No compute

Three patterns cover this design

- **On-prem: a Universal Agent per data center.** A single agent with only the capabilities each site needs toggled on in the portal. Devices point at it, and it talks to Kentik over outbound HTTPS. The standalone kproxy agent is not used; the Universal Agent Flow Proxy replaces it.
- **Public cloud: agentless.** Flow logs go to storage in the customer's own account, and Kentik gets read-only access to the logs, topology metadata and metrics.
- **Internet and SaaS: synthetic tests.** Kentik's global agents and the customer's own agents test SaaS apps, websites, DNS and paths between sites and clouds.

Source: Kentik KB, Universal Agents; Public Clouds; Synthetics Overview

2. Topology


Main Data Center · Denver, CO
DATA CENTER
Network devices
Routers, switches, firewalls
Servers (syslog)
Universal Agent
Docker host
Flow Proxy
SNMP/ST
Trap Receiver
Syslog Server
Connectivity Test
Internet & SaaS apps
Synthetic test targets
Tests: Epic Hosting, Microsoft 365
Kentik global agents: 2
Test results
HTTPS, outbound
Kentik SaaS
Customer's Kentik account
United States (Ashburn, VA)
Ingest and enrichment
Kentik Data Engine (KDE)
Data Explorer, Kentik Map
NMS, Synthetics, Cloud, alerts
AI Advisor and APIs
Kentik reads, read-only
AWS
AGENTLESS
VPC, TGW and firewall flow logs
S3 bucket per region
Read-only IAM role
Topology metadata (required)
VPC and Transit Gateway flow logs
To Kentik over HTTPS
Kentik reads cloud logs and metadata
Synthetic tests
Only outbound connections from customer networks. No inbound access from the Internet.

3. Roles and permissions

Who on the customer side does what, and the access each task needs. Kentik only ever receives read-only access.

Role	Task	Access needed
Kentik administrator	Deploy and authorize Universal Agents, enable capabilities, add devices, create cloud exports and tests	Kentik admin (or super admin) user in the customer's Kentik account
Server or platform team	Provide 1 agent host(s) and run the install command	root or sudo on a supported Linux host, or Docker
Network team	Point flow, traps and syslog at the agents; create SNMP and SSH credentials	Device configuration rights; SNMP v2c/v3 read-only credentials; read-only SSH account
Security / firewall team	Open the rules in section 4	Firewall and proxy change rights
AWS IAM administrator	AWS: enable flow logs and grant Kentik access	IAM role trusting arn:aws:iam::834693425129:root with sts:ExternalId = Kentik Company ID; Describe/List APIs, CloudWatch read, s3:Get* and s3:List* on the log bucket

Access granted to Kentik

Scope	Access
AWS	IAM role trusting arn:aws:iam::834693425129:root with sts:ExternalId = Kentik Company ID; Describe/List APIs, CloudWatch read, s3:Get* and s3:List* on the log bucket
Network devices	SNMP read-only (polled from the customer's own agent, never from the Internet)

Source: Kentik KB, UA Deployment; NMS Setup; SSH Command Access for Kentik NMS; Kentik for AWS

4. Network and firewall requirements

All Universal Agent connections are outbound. Nothing needs to reach into the customer network from the Internet.

Outbound allowlist (every agent)

Destination	Port	Notes
flow.kentik.com	TCP 443	TLS 1.2+
grpc.api.kentik.com	TCP 443	TLS 1.2+
api.kentik.com	TCP 443	TLS 1.2+
portal.kentik.com	TCP 443	TLS 1.2+
storage.googleapis.com	TCP 443	TLS 1.2+
whoami.kentiklabs.com	TCP 443	Optional
Kentik flow ingest	UDP 40010	Optional Flow Proxy base port

- SSL-inspecting firewalls and proxies must allow gRPC over port 443.
- UDP 40010 to Kentik is an optional Flow Proxy base port.
- A web proxy works: set https_proxy before running the install (Linux) or pass it with --env (Docker).

Main Data Center · Denver, CO

From	To	Port	Purpose
Universal Agent	Kentik SaaS (flow.kentik.com, grpc.api.kentik.com, api.kentik.com, portal.kentik.com, storage.googleapis.com)	TCP 443, TLS 1.2+	All telemetry, control and updates (outbound only)
Universal Agent	Kentik SaaS	UDP 40010 (optional)	Flow Proxy base port
Network devices	Universal Agent	UDP 9995	Flow export (NetFlow, sFlow, IPFIX)
Universal Agent	Network devices	UDP 161	SNMP polling (v1, v2c, v3)
Universal Agent	Network devices	ICMP	Reachability
Universal Agent	Network devices	Device gNMI port	Streaming telemetry dial-in (only if used)
Network devices	Universal Agent	UDP 162	SNMP traps
Network devices and servers	Universal Agent	UDP and TCP 514	Syslog
Other Universal Agents	Universal Agent (both directions)	TCP 8877 and UDP 9977	Agent-to-agent and mesh tests
Universal Agent	Test targets	ICMP, TCP or UDP	Ping and traceroute
Universal Agent	Test targets	TCP 80 and 443	HTTP(S) and API tests

Source: Kentik KB, Universal Agents; UA Deployment; Test Types Reference

5. Universal Agent deployment

Install one Universal Agent per site, then switch capabilities on in the portal. The agent updates itself and reports its own health.

Host requirements

- Supported OS: Debian 11, 12, 13; Ubuntu 20.04, 22.04, 24.04; Oracle Linux 9; CentOS Stream 8. CentOS 7 and Enterprise Linux before 8 are not supported.
- Or any Docker host that can run the generated docker run command.
- Size for the sum of the capabilities on the host, plus 20 to 30% headroom for bursty flow, syslog and traps. The plans below use 30%.
- Disk sizing is not published in the KB; confirm with Kentik.

Source: Kentik KB, Universal Agents; UA Deployment

5.1 Main Data Center · Denver, CO

Data center. Install method: **Docker**. Capabilities: **Flow Proxy: Forwarding, SNMP/ST, SNMP Trap Receiver, Syslog Server, Connectivity Test**.

Capability	Planned load	CPU / RAM
Flow Proxy (Edge Processing)	10,000 flows/s	8 cores / 32 GB
SNMP/ST	120 devices	2 cores / 8 GB
Syslog Server	1,000 events/s	1 cores / 1 GB
Connectivity Test	per agent	2 cores / 2 GB
SNMP Trap Receiver	event driven	Not published
Total from KB sizing		13 cores / 43 GB
Recommended host (+30%)		17 vCPU / 56 GB RAM

Install and register

1. In the Kentik portal go to Settings > Universal Agents and click + Deploy Agent.
2. Under Installation pick Docker. The portal generates a command that carries K_COMPANY_ID and K_REGISTER_PROVISIONING_TOKEN.
3. Copy the command and run it on the agent host. Portal tokens last one hour, and you can register as many agents as you like in that window.
4. The agent appears under Select an Agent (host name is the container ID). Click Authorize. An agent left unauthorized shows Not Authorized and does nothing.
5. Set Display Name, Description and Site Name (Main Data Center), then click Continue.
6. Toggle on the capabilities for this site (Flow Proxy: Forwarding, SNMP/ST, SNMP Trap Receiver, Syslog Server, Connectivity Test) and click Finish.
7. Confirm the agent status is Up. An agent not seen for 30 minutes shows Down. Agents update themselves and their capabilities.

```
# run the docker command copied from Settings > Universal Agents > Deploy Agent
# it carries K_COMPANY_ID and K_REGISTER_PROVISIONING_TOKEN
```

Source: Kentik KB, UA Deployment; UA Portal Management; Device Settings

6. Capability configuration

How to configure each capability used in this design, on the agent and on the devices that feed it.

Flow Proxy: Forwarding

Receives NetFlow, sFlow and IPFIX from routers, switches and firewalls, enriches it and forwards it to Kentik over TLS.

Enabled at: Main Data Center · **License:** FlowPak

- Toggle on Flow Proxy: Forwarding on the agent. Settings: Host 0.0.0.0, Port 9995 (UDP), optional sFlow agent address override, and Site.
- On each device, export flow to the agent's IP on UDP 9995. Sample ingress on every interface; never ingress and egress on the same interface.
- The sample rate on the device must match the rate set for that device in Kentik.
- Vendor configuration snippets for flow, SNMP and BGP: github.com/kentik/config-snippets.
- Sanity check after go-live: flow volume should land within about 20% of SNMP interface traffic.

Source: Kentik KB, Universal Agents; Router Configuration; Flow Overview; Migration Scenarios (kproxy to UA)

SNMP/ST

Discovers devices and collects health, interface and hardware metrics by SNMP polling or gNMI streaming telemetry. It also enriches flow with interface names and speeds.

Enabled at: Main Data Center · **License:** FlowPak (flow enrichment) or NMS / DevicePak (full monitoring)

- Run the NMS Setup Wizard: pick this agent, enter IP ranges and SNMP v1, v2c or v3 credentials, discover, then add devices.
- Per device, on the SNMP tab choose "Flow Enrichment via Universal Agent" or "Full Monitoring via Universal Agent".
- SNMP v3 options: auth None, MD5 or SHA; privacy None, DES or AES.
- Default polling: 1 minute for status, CPU, memory and interfaces; 5 minutes for everything else.
- Streaming telemetry (optional): enable "Dial-In from Universal Agent" (gNMI over gRPC, 30 second default).

Source: Kentik KB, Universal Agents; NMS Setup; Device Settings

SNMP Trap Receiver

Captures SNMP traps the moment a device sends them, for example an interface or BGP session going down.

Enabled at: Main Data Center · **License:** NMS / DevicePak

- Enable SNMP Trap Receiver on the agent.
- Point device trap targets at the agent's IP on UDP 162.
- Traps must come from a device licensed for NMS. Query them in Data Explorer with the metric "SNMP Traps: Events".

Source: Kentik KB, Universal Agents; SNMP Traps & Syslog

Syslog Server

Collects log messages from devices and servers so they can be searched, alerted on and lined up with traffic.

Enabled at: Main Data Center · **License:** NMS / DevicePak

- Enable Syslog Server on the agent.

- Point device and server syslog at the agent's IP on UDP or TCP 514.
- If a syslog forwarder sits in between, it must keep the original device IP. Changing the port currently needs Kentik support.
- Query in Data Explorer with the metric "Syslog: Events".

Source: Kentik KB, Universal Agents; SNMP Traps & Syslog

Connectivity Test

Makes this agent a private test point. It measures latency, loss and jitter to other sites, clouds, SaaS apps and DNS servers, and gives AI Advisor on-demand ping and traceroute.

Enabled at: Main Data Center · **License:** Synthetics test credits

- Toggle on Connectivity Test on the agent.
- Open TCP 8877 and UDP 9977 between every pair of agents that will test each other.
- Create tests in Synthetics: agent-to-agent, network mesh, server IP or hostname, DNS server monitor and HTTP(S) or API tests.
- Confirm which continuous test types run on the Universal Agent today with the Kentik account team.

Source: Kentik KB, Universal Agents; Synthetics Overview; Test Types Reference; Test Credits

7. Public cloud setup

No agents are needed in public clouds. Each cloud writes flow logs to storage in the customer's own account, and Kentik reads them with read-only access. Kentik recommends a metadata-only export first to prove access before turning on high-volume logs.

Amazon Web Services

No agent. Flow logs are written to S3 buckets the customer owns, and Kentik assumes a read-only IAM role protected by an ExternalId.

Collecting: Topology metadata (required), VPC and Transit Gateway flow logs. Regions: us-east-1.

Steps

1. Start with a metadata-only export to prove access before turning on high-volume logs.
2. Create one S3 bucket per region for flow logs. Kentik treats each bucket as one device, so hundreds of VPCs into one global bucket risks S3 throttling.
3. Encryption: SSE-S3 is recommended. SSE-KMS needs the KMS statement. SSE-C is not supported.
4. Turn on flow logs for VPCs, Transit Gateways: filter All, destination S3, custom format with every v2 to v5 field.
5. Create the IAM role with the trust policy and permissions below.
6. In Kentik: Settings > Public Clouds > Create Cloud Export > AWS Cloud. Select features, enter the Role ARN > Verify Role, Region > Verify Region, bucket name and prefix > Verify S3 Bucket.
7. Pick sampling (Legacy, a rate from 2 to 2000, or Unsampled), name the export, choose the billing plan and Save.
8. Optional: tick "Help me configure my provider via Terraform" for a main.tf that creates the role, bucket and flow logs.

IAM role trust policy

```
{
  "Version": "2012-10-17",
  "Statement": [{
    "Effect": "Allow",
    "Principal": { "AWS": "arn:aws:iam::834693425129:root" },
    "Action": "sts:AssumeRole",
    "Condition": {
      "StringEquals": { "sts:ExternalId": "<Kentik Company ID>" },
      "ArnEquals": { "aws:PrincipalArn": "arn:aws:iam::834693425129:role/eks-ingest-node" }
    }
  }]
}
```

IAM permissions policy

Service	Actions
EC2	ec2:Describe*, ec2:Search*, ec2:GetManagedPrefixListEntries
Direct Connect	directconnect:List*, directconnect:Describe*
ELB	elasticloadbalancing:Describe*
IAM	iam:ListAccountAliases
Network Firewall	network-firewall:Describe*, network-firewall:List*

Service	Actions
Cloud WAN	networkmanager:ListCoreNetworks, GetCoreNetwork, GetCoreNetworkPolicy, ListAttachments, GetNetworkRoutes
CloudWatch	cloudwatch:ListMetrics, GetMetricStatistics, GetMetricData, Describe*
Organizations	organizations:ListAccounts
S3 (log bucket)	s3:Get*, s3:List* on arn:aws:s3:::<bucket> and arn:aws:s3:::<bucket>/*
KMS (SSE-KMS only)	kms:Decrypt, kms:DescribeKey on the bucket key

Export status values: Start, Pending, Error, OK, Disabled or Halt. Licensing: CloudPak, measured in flows per second.

Source: Kentik KB, Kentik for AWS; Metadata Configuration (AWS); Flow/Firewall Log Collection Configuration (AWS); Kentik Export Configuration for AWS; Public Clouds

8. Synthetic monitoring plan

Continuous tests measure latency, loss and jitter between sites, clouds and the applications users depend on.

Test points

Agent	Type
Main Data Center	Private (Universal Agent)
2 Kentik global agent(s)	Kentik-hosted: Internet hubs, cloud regions, app agents

Planned tests

- HTTP(S) test to Epic Hosting from private and global agents.
- HTTP(S) test to Microsoft 365 from private and global agents.
- Available test types: Agent-to-Agent, Network Mesh, Server IP, Server Hostname, Network Grid, autonomous tests (ASN, CDN, country, region, city), DNS Server Monitor, DNS Server Grid, HTTP(S) or API, Page Load and Transaction.

Estimated monthly test credits

Test	Interval	Per run	Credits / month
HTTP(S) tests to 2 SaaS app(s)	every 5 min	1 private + 2 global agents, 5 s timeout = 25 per run	216,000
Total			216,000

Credit rules from the KB: network subtest 2.5 credits from a private agent and 5 from a global agent; HTTP subtests cost timeout (seconds) x 0.5 private or x 1 global; BGP Monitor 0.53 credits per minute; a test every minute runs 43,200 times in a 30-day month; Test Preview is free until saved. Mesh estimates count each direction as one subtest (the KB example: a 4-agent private mesh is 30 credits per run). Treat these as planning estimates and confirm with Kentik.

Source: Kentik KB, Synthetics Overview; Test Types Reference; Test Credits; Agents Page

9. Go-live validation checklist

☐	Main Data Center: agent shows Up in Settings > Universal Agents, capabilities show enabled
☐	Main Data Center: flow arrives in Data Explorer; device sample rates match Kentik; flow within about 20% of SNMP interface traffic
☐	Main Data Center: devices discovered in NMS with interfaces and health metrics
☐	Main Data Center: test trap or syslog message visible (SNMP Traps: Events, Syslog: Events)
☐	AWS: cloud export status OK; metadata visible in Kentik Map; flow logs arriving
☐	Synthetics: tests show Healthy, Warning or Critical (not Pending); agents Active
☐	Alerting: at least one alert policy routed to the customer's notification channel

Source: Kentik KB, UA Portal Management; Flow Overview; SNMP Traps & Syslog; Public Clouds

10. Security, data handling and licensing

Area	Detail
Hosting	US: Ashburn, VA. EU: Frankfurt. Chosen when the contract is signed.
In transit	TLS 1.2+ to and from Kentik
At rest	AES-256, consistent with FIPS 140-2
Isolation	Separate logical database and tables per customer
Audits	SOC 2 Type II yearly, annual third-party penetration test
Access	SAML 2.0 SSO, MFA (TOTP or hardware keys), user, admin and super admin roles, per-user API tokens
Data	IP addresses and network metadata, never packet payloads; IP redaction available
Retention (default)	45 days full-resolution flow, 120 days summarized, 3 or 13 months NMS metrics by plan; contracts can change this

Source: Kentik KB, Security Overview; KDE Overview

Licensing

License	Covers
FlowPak	Flow from routers, switches, firewalls and hosts, including SNMP flow enrichment.
NMS / DevicePak	Full device monitoring, plus syslog, SNMP traps and SSH config and commands.
CloudPak	Public cloud flow logs, sized in flows per second.
Synthetics	Monthly test credits: global agent subtest 5 credits, private agent 2.5, BGP Monitor 0.53 per minute.

Plans and editions vary by contract. Confirm entitlements with the Kentik account team.

Source: Kentik KB, Licenses; Test Credits

11. Open items to confirm with Kentik

- Which continuous synthetic test types run on the Universal Agent today (the KB names the capability Connectivity Test).
- Licensing names and entitlements per contract (FlowPak, NMS/DevicePak, CloudPak, Synthetics credits).
- Disk sizing for Universal Agent hosts is not published in the KB.
- BGP Proxy shows in Installed Capabilities in the portal, but the KB page still lists it as coming soon. Confirm it is available for this account before relying on it.
- Flow Proxy: Forwarding is marked beta in the KB. Confirm the flavor (Forwarding or Edge Processing) per site.

12. Appendix: Kentik KB references

Page	URL
Universal Agents	https://kb.kentik.com/docs/universal-agents
Public Clouds	https://kb.kentik.com/docs/public-clouds
Synthetics Overview	https://kb.kentik.com/docs/synthetics-overview
UA Deployment	https://kb.kentik.com/docs/ua-deployment
NMS Setup	https://kb.kentik.com/docs/nms-setup
SSH Command Access for Kentik NMS	https://kb.kentik.com/docs/ssh-command-access-for-kentik-nms
Kentik for AWS	https://kb.kentik.com/docs/kentik-for-aws
Test Types Reference	https://kb.kentik.com/docs/test-types
UA Portal Management	https://kb.kentik.com/docs/ua-portal-management
Device Settings	https://kb.kentik.com/docs/device-settings
Router Configuration	https://kb.kentik.com/docs/router-configuration
Flow Overview	https://kb.kentik.com/docs/flow-overview
Migration Scenarios (kproxy to UA)	https://kb.kentik.com/docs/kproxy-migration-scenarios
SNMP Traps & Syslog	https://kb.kentik.com/docs/snmp-traps-syslogs
Test Credits	https://kb.kentik.com/docs/test-credits
Metadata Configuration (AWS)	https://kb.kentik.com/docs/metadata-configuration-aws
Flow/Firewall Log Collection Configuration (AWS)	https://kb.kentik.com/docs/flow-firewall-log-collection-configuration
Kentik Export Configuration for AWS	https://kb.kentik.com/docs/kentik-export-configuration-aws
Agents Page	https://kb.kentik.com/docs/agents-page
Security Overview	https://kb.kentik.com/docs/security-overview
KDE Overview	https://kb.kentik.com/docs/kde-overview
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